Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Foundation of Data Science and Analytics (CSUE203) (Elective-I)

Semester: 4th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	4	2	-	6	5
Marks	100	50	-	150	

Course Pre-requisites:

- Programming language and Database Management System

Course Description:

This course offers a fundamental understanding of the concepts, tools, and techniques in Data Science and Analytics. Students will explore various stages of data collection, pre-processing, exploratory data analysis (EDA), visualization, and analytics. Through theoretical discussions and practical applications, they will learn to handle different types of data, employ data pre-processing methods, and apply visualization techniques for insightful decision-making including the predictive analysis. The course also emphasizes ethical issues and real-world problem-solving across various domains.

Course Objectives:

1. To develop students' proficiency in data collection techniques and pre-processing methods for real-world applications.
2. To demonstrate different statistical methods for Exploratory Data Analysis (EDA).
3. To develop problem solutions using data analytics techniques with appropriate visualizations and interpreting results through effective storytelling with data.

Course Outcomes:

By the end of the course, students will be able to:

1. **CO1**: Apply pre-processing techniques on data and prepare it for analysis.
2. **CO2**: Conduct exploratory data analysis using statistical methods and feature engineering techniques.
3. **CO3**: Apply appropriate data visualization methods and create dashboards that communicate insights clearly.
4. **CO4**: Solve real-world problems by applying analytics techniques, including descriptive, diagnostic, predictive, and prescriptive models.

- **Pre-requisites online video / course:**
  - Introduction to R: https://www.udemy.com/course/introduction-to-r/

- **Self-study components and materials:**

Unit No.	Unit/Topic Details	
1	**INTRODUCTION**	
	1.1	Introduction to Data Science and Data Analytics
	1.2	Applications and Emerging Trends of Data Science
	1.3	Ethical and legal issues in data science
	1.4	Data collection techniques – Primary and Secondary Data Collection and types of data
	1.5	Data Cleaning - Missing Values, Noisy Data, Data Cleaning as a Process
	Materials: ➢ https://www.ibm.com/topics/data-science ➢ https://www.youtube.com/watch?v=RBSUwFGa6Fk ➢ https://www.microsoft.com/en-us/ai/responsible-ai ➢ https://www.coursera.org/articles/data-collection-methods ➢ https://www.splunk.com/en_us/blog/learn/data-structured-vs-unstructured-vs-semi-structured.html ➢ "Data Mining Concepts and Techniques", by Jiawei Han, Micheline Kamber, Jian Pei, Third Edition (pp. 88 to 91)	
2	**EXPLORATORY DATA ANALYTICS (EDA)**	
	2.1	Descriptive statistics – Mean, Median, Mode, Range, Standard Deviation
	Materials: ➢ https://www.youtube.com/watch?v=SplCk-t1BeA	
3	**DATA VISUALIZATION**	
	3.1	Importance of Data Visualization
	3.2	Types of Visualizations
	Materials: ➢ https://www.coursera.org/articles/data-visualization ➢ How to turn data into stories by Cole Nussbaumer Knaflic | Talks at Google, https://www.youtube.com/watch?v=Hfx1X9WSGYQ ➢ "The Truthful Art: Data, Charts, and Maps for Communication", by Alberto Cairo (Read pp. 41-65 & 121- 149)	
4	**DATA ANALYTICS**	
	4.1	Importance of Data Analytics
	Materials: ➢ https://online.hbs.edu/blog/post/types-of-data-analysis	

Syllabus:

Unit No.	Unit/Topic Details		Hours (hr)	Evaluation Weightage (%)
1	**OVERVIEW AND DATA PRE-PROCESSING**		12	20
	1.1	Difference between Data analysis and analytics		
	1.2	Data Science Lifecycle		
	1.3	Roles and Challenges in Data Science		
	1.4	Types of data attributes		
	1.5	Major tasks in Data Pre-processing		

	1.6	Data Integration - Entity Identification Problem, Tuple Duplication		
	1.7	Data Transformation – Min-max normalization, z-score normalization, and Decimal scaling; Binning		
	1.8	Data encoding – Label encoding and one-hot encoding		
	1.9	**Pre-processing** Semi-structured data - Parsing and Extraction, Converting semi-structured data into structured formats Unstructured data - Text: Text Cleaning, Stopword Removal and Tokenization; Image: Image Resizing, Cropping, Histogram Equalization, Noise Reduction, and Normalization		
2	**EXPLORATORY DATA ANALYTICS (EDA)**		18	30
	2.1	Hypothesis Testing		
	2.2	Data Sampling - Simple Random Sampling, Systematic Sampling, Stratified Sampling, Stratified Random Sampling, Cluster sampling		
	2.3	Univariate, bivariate, and overview of multivariate analysis		
	2.4	Introduction to ANOVA, Two-way ANOVA		
	2.5	Feature selection - Filter Methods: Correlation and chi-square, Wrapper Methods: Recursive Feature Elimination, Forward/Backward Selection Embedded Methods		
	2.6	Feature Extraction - Principal Components Analysis		
	2.7	Regression analysis – Simple and Multiple		
	2.8	Outliers and Anomalies		
	2.9	Text data: Bag of Words, Word cloud Image data: Image Augmentation, Edge Detection and Contour Analysis		
3	**DATA VISUALIZATION**		12	20
	3.1	Principles of Effective Visualization		
	3.2	Chart Selection Strategies		
	3.3	Multivariate Data Visualization - Scatter Plot Matrix, Heatmap, Bubble Plot, Parallel Coordinate Plots		
	3.4	High-Dimensional Data Visualization - t-SNE		
	3.5	Geospatial Data Visualization - Choropleth Maps, Dot Density Maps, Contour Maps, 3D Maps, Topographic map, Cartogram map, Hexagonal binning map		
	3.6	Time Series Visualization - Line Chart, Candlestick Chart		
	3.7	Interactive Visualizations: Interactive Features and Charts, Real-Time Data Integration, Dashboard tool - Google looker studio & Tableau		
	3.8	Storytelling with Data		
4	**DATA ANALYTICS**		18	25
	4.1	Types of Data Analytics		
	4.2	Descriptive Analytics & Diagnostic Analytics - Root Cause Analysis, Drill-Down Analysis		
	4.3	Predictive Analytics: Predictive Modeling – Classification using Logistic regression and k-nearest neighbors, clustering using k-means; Time-Series Analysis – Moving average, ARIMA model		

	4.4	Prescriptive Analytics - Optimization Techniques – Types of optimization models, Linear programming problem, Non-linear programming problem - Gradient descent; Simulation Modeling - Fundamentals of simulation, simulation types, Monte-Carlo simulation		
			Total Hours:	**60**

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	3	2	-	-	-	-	-	-	2	-	-
CO2	3	3	3	3	2	-	-	-	-	-	-	2	-	-
CO3	2	3	3	3	3	-	-	-	-	3	-	3	-	-
CO4	3	3	3	3	3	-	-	-	-	-	-	3	-	-

The correlation levels 1, 2 or 3 are as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), If there is no correlation, put "-" placed.

Recommended Study Material:

- **Text books:**

  1. "Data Mining Concepts and Techniques", by Jiawei Han, Micheline Kamber, Jian Pei, Third Edition, Elsevier, 2012. ISBN 978-0-12-381479-1
  2. "R in Action: Data Analysis and Graphics with R", by Robert I. Kabacoff, Manning, 2nd Edition

- **Reference Books:**

  1. "Beginning Data Science in R: Data Analysis, Visualization, and Modelling for the Data Scientist", by Thomas Mailund, Apress publishing
  2. Wu, J., and Coggeshall, S. Foundations of Predictive Analytics. CRC Press.
  3. "Prescriptive Analytics - The Final Frontier for Evidence-Based Management and Optimal Decision Making", by Dursun Delen, Pearson

- **Online resources:**

  1. Advanced R Programming for Data Analytics in Business, by Prof. A. Tripathi, IIT Kanpur, https://onlinecourses.nptel.ac.in/noc24_mg113/preview
  2. Practitioners Course In Descriptive, Predictive And Prescriptive Analytics , by Prof. D. Philip and Prof. A. Oberoi, IIT Kanpur https://onlinecourses.nptel.ac.in/noc19_mg42/preview
  3. Coursera, Introduction to R Programming for Data Science, https://www.coursera.org/learn/introducton-r-programming-data-science

- **Tools:**

  - RStudio, Orange, Google looker studio, Power BI, and Tableau